SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	THEJAS.B	**Reg. No.:**	192565092
Section / Batch:	2025	**Date:**	24/07/2026

Code of Conduct

I THEJAS.B /192565092 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: THEJAS . B

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____________	Signature: _____________	Signature: _____________
Date: _________________	Date: _________________	Date: _________________

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low <= high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

Q2 **Stack** 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop(){
    if(top==-1){
        printf("underflow");
        return-1;
    }
    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

Run Save Submit Fix

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

Q3 **Stack**

5 marks

Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int stack[5];
int top;
void init(){
    top=-1;
}
```

Test Input

stdin input (optional)

Output

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Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch == ')') {
    while(top != -1 && stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- **Q1** Binary Search 5 marks
- **Q2** Stack 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Stack - Balancing Symbols 5 marks
- **Q6** Singly Linked List 5 marks
- **Q7** Stack 5 marks
- **Q8** Linked List 5 marks
- **Q9** Linked List 5 marks
- **Q10** Queue 5 marks

Q5 **Stack - Balancing Symbols** 5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(top!=-1&&(
    (ch == ')' && stack[top] == '(') ||
    (ch == '}' && stack[top] == '{') ||
    (ch == ']' && stack[top] == '[')
)) {
    pop();
}else{
    return 0;
}
```

Test Input

stdin input (optional)

Output

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Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Q6 **Singly Linked List**

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
temp=head
while temp is not none:
    print(temp.data)
    temp=temp.next
```

Test Input

stdin input (optional)

Output

Run Save Submit Fix

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- Q1 Binary Search 5 marks
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- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
def POP(stack, top):
    if top < 0:
        print("Underflow")
        return top  # or return None depending on your design
    value = stack[top]
    top -= 1
    return value, top
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

Q8 **Linked List** 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
prev = None
current = head

while current is not None:
    next_node = current.next
    current.next = prev
    prev = current
    current = next_node

head = prev
```

Test Input

stdin input (optional)

Output

Run Save Submit Fix

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QUESTIONS

- Q1 Binary Search 5 marks
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- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

Q9 **Linked List** 5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
// assume new is a newly created node
new->next = head;
head = new;
```

Test Input

stdin input (optional)

Output

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Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
DEQUEUE(Q):
if front == -1 or front > rear:
    print("Underflow")
else:
    x = Q[front]
    front = front + 1
    if front > rear:
        front = -1
        rear = -1
    return x
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192565092RA
_6a7027a42674b7.69118420\main.c:1:1
: warning: return type defaults to
'int' [-Wimplicit-int]
```

Pending manual review

Submitted